

5. Failures of the CAPM

The CAPM is derived using some very strong assumptions. It's worth taking a moment to examine these assumptions and discuss what happens when they are relaxed.

1. Investors have only financial wealth.

As Part I of this book has emphasized, investors have unique income streams and liabilities, and their optimal portfolio choice has to take these into consideration. Liabilities are often denominated in real terms—we want to maintain a standard of living even if prices rise, for example. Income streams are usually risky, and income declines during periods of low economic growth. This makes variables like inflation and growth important factors because many investors' income and liabilities change as the macro variables change.

One particular important factor that drives asset returns is human capital, or labor income risk.¹⁰ In an influential paper, Jagannathan and Wang (1996) found large improvements in the performance of the CAPM when labor income risk is taken into account.

2. Investors have mean-variance utility.

As chapter 2 emphasizes, more realistic utility functions often have an asymmetric treatment of risk because investors are generally more distressed by losses than pleased by gains. We should expect, then, to find deviations from the CAPM among stocks that have different measures of downside risk. Ang, Chen, and Xing (2006) show that stocks with greater downside risk have higher returns. A large number of papers show that other higher moment risk, like skewness and kurtosis, also carry risk premiums.¹¹

3. Single-period investment horizon.

By itself an investment horizon of one period is a minor assumption. Merton (1971, 1973) provides a famous extension of the CAPM to the dynamic case. In this setting, the CAPM holds in each single period.

While the long investment horizon is an inconsequential assumption for the CAPM theory, there is a huge implication when we extend portfolio choice to a dynamic, long-horizon setting. As Part I of this book has shown, the optimal strategy for long-term investors is to rebalance (see chapter 3).

¹⁰ Mayers (1973) is the seminal first reference. See also Constantinides and Duffie (1996), Jagannathan, Kubota, and Takehara (1998), Storesletten, Telmer, and Yaron (2007), and Eiling (2013).

¹¹ These effects come in two forms. First, there is the risk premium associated with individual stock higher moments. These are properties of each individual stock. See Mitton and Vorkink (2007), Boyer, Mitton, and Vorkink (2010), and Amaya et al. (2012) for skewness risk premiums of this form. Second, there is the risk premium coming from how stock returns covary with higher moments of the aggregate market. Harvey and Siddique (2000), Dittmar (2002), and Chang, Christoffersen, and Jacobs (2013) show that there are risk premiums for co-skewness and co-kurtosis, which result from the co-movement of stock returns with skewness and kurtosis moments of the market portfolio.